ICLAB 2017Fall Midterm Exam

Name: **Student ID:**

1. [15%] Please answer the following questions:

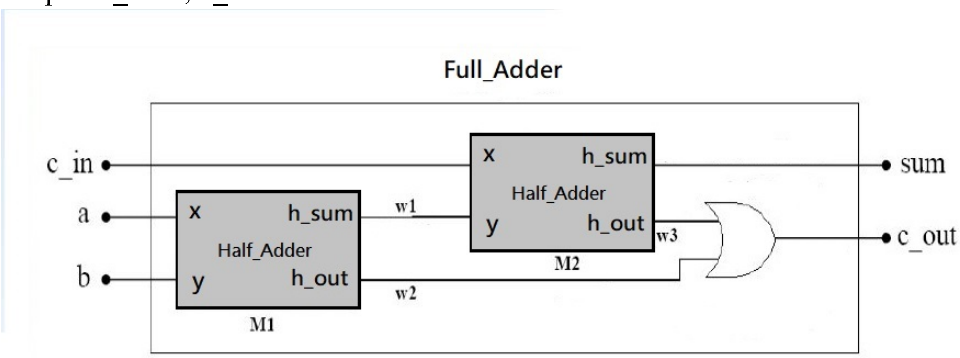
   a. [5%] Why we need different delay time Mid/Typ/Max for our synthesis?

   b. [5%] What is the different between "==' and '==='in Verilog?

   c. [5%] Why don't we use array as memory? (Please describe two major reasons)

2. [10%] Write down the following circuit scheme by calling the module "Half_Adder". Please follow the name list.

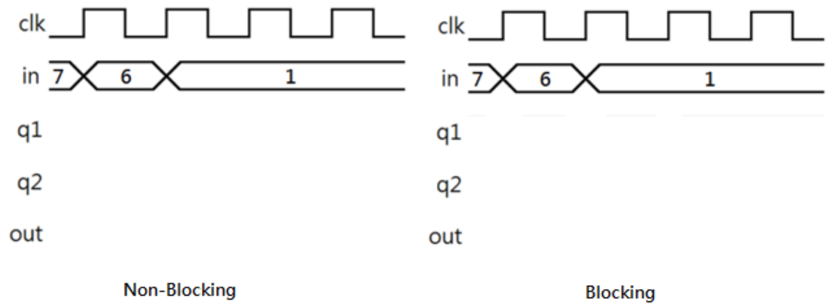
   Module1 name: Full_Adder
   Input : a , b , c_in
   Output: sum , c_out
   Module2 name: Half_Adder
   Input : x , y
   Output: h_sum , h_out



3. [10%] Please calculate the value of each node in different clock cycle for non-blocking and blocking.

```
always @(posedge clk)
begin
   q1 <= in;
   q2 <= q1;
   out <= q2;
end
```

```
always @(posedge clk)
begin
   q1 = in;
   q2 = q1;
   out = q2;
end
```



4. [5%] Please describe the differences between Moore machine and Mealy machine.

5. [5%] Please simply describe their roles in design environment (PATTERN.v, TESTBENCH.v, RTL.v)
6. [5%] Please list at least two differences between task and function
7. [10%] Timescale is very important in hardware design, so it is necessary to set the right timescale while designing your architecture.

```
`timescale 1ns/XXXps
module TEST
    parameter CYCLE = 2.45;
    reg     CLK;
    initial   CLK = 0;
    always #(CYCLE/2.0) CLK = ~CLK;
endmodule
```

   a. [5%] Try to explain the two parts (1ns/XXXps) of command `timescale
   b. [5%] What is the value of XXX for precision requirement?

8. [10%] Timing
   a. [5%] Please briefly explain the term **skew time**.
   b. [5%] How to fix **setup time violation** and **hold time violation**?
9. [5%] Please list three kinds of **IP (Intellectual Property)** and briefly explain their **difference (two** features for each one**)**.

10. [5%] Please state the main reason why hardware designers prefer to use SRAMs instead of registers to implement large storage. Also, if we would like to storage 1500*16-bit data in a SRAM which has the same architecture in Lab5, how many input pins and address pins do we need? List one possible solution.

11. [10%] The following module is an image processor. **CONV_IM** is a large combinational circuit and it will convolute the image **im** with the filter **f** then store the results in the SRAM RA1SH. The **wen_sram** and the address **addr** will control the module in write mode or read mode. **POST_IM** is a large combinational circuit and function as a post image processor. Assuming the functionality of the submodules **CONV_IM** and **POST_IM** are correct. What are the potential problems in the Verilog code below? Please propose some solutions to fix those problems.

```
module image_processor (clk, im, f, addr, wen_sram, post_out) begin
input [1023:0] im;
input [5:0] f;
input [1023:0] addr;
input wen_sram;
output [1023:0] post_out;
wire [1023:0] im_out;

CONV_IM conv_im1( .output(im_out), .image(im), .filter(f) ) ;
RA1SH U1(.Q(sram_out),
            .CLK(clk),
            .CEN(1'b0),
            .WEN(wen_sram),
            .A(addr),
            .D(im_out),
            .OEN(1'b0));
POST_IM post_im1(.output(post_out), .input(sram_out));
end
```

12. [10%] Optimize the design

   a. [5%] Please list three methods to resolve **multiple instances** before running the compile command.

   b. [5%] Among synthesis algorithm, using **compile_ultra** command can maximize performance and minimize area. However, when will designer choose to use **compile** command instead of **compile_ultra**?